Homework #2

This is the homework for week 3. The due date is 11:59 p.m., September 22. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (20 Points) Suppose $X_1, X_2, \cdots, X_{50}$ are i.i.d. Poisson distributed random variables with parameter $\lambda = 3$. Approximate the probability of $\sum_{i=1}^{50} X_i \in [1, 10]$ using central limit theorem.

(**Question2):** (20 Points) Let Y be $\chi^2(n)$ distributed random variable. Derive the limiting distribution of

$$W = (Y - n)/\sqrt{2n} \text{ as } n \to \infty.$$

(**Question3):** (40 Points) Suppose $Z_1, Z_2, \cdots, Z_n$ are i.i.d. Poisson distributed random variables with parameter λ.

i. Suppose A and B are independent Poisson distributed random variables with parameter λ_1 and λ_2. Prove that

$$A + B \text{ satisfies Poisson distribution with parameter } \lambda_1 + \lambda_2.$$

ii. Use induction to show that $\sum_{i=1}^{n} Z_i$ satisfies Poisson distribution with parameter $n\lambda$.

iii. Derive the asymptotic distribution of

$$\sqrt{n}(\frac{1}{n}\sum_{i=1}^{n} Z_i - \lambda).$$

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. For the sample size $n = 5, 10, 20, 50$, generate $Z_i, i = 1, 2, \cdots, n$ Poisson distributed random variables with parameter 1. Calculate $A = \frac{1}{n}\sum_{i=1}^{n} Z_i$ and $B = \sqrt{n}(A - 1)$.

ii. Repeat step 1 for 200 times, plot histograms of A and B. Overall, you should demonstrate 8 histograms, 4 for A and 4 for B.